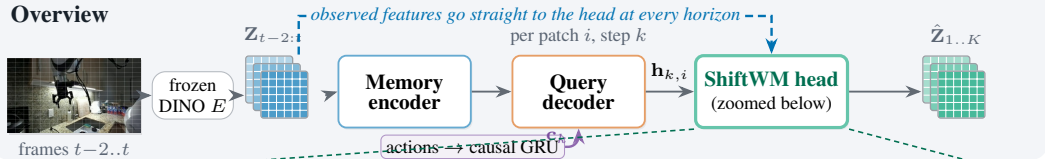


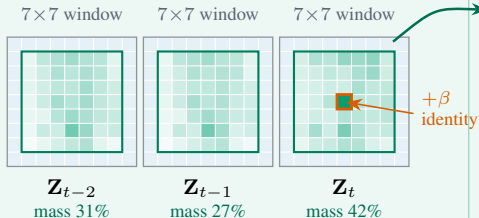
Overview



Inside the ShiftWM head: one patch i at horizon $k=10$ (trained model, held-out DROID)

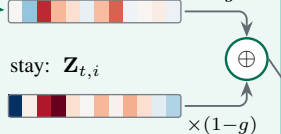
1 Where does patch i come from?

$$\pi_{k,i}(j) = \text{softmax}_{j \in \mathcal{W}(i)} (\mathbf{q}_{k,i}^\top \mathbf{k}_j / \sqrt{d} + \beta \mathbf{1}[j=(t,i)])$$



2 Move, or stay

$$\text{move: } \mathbf{T}_{k,i} = \sum_j \pi_{k,i}(j) \mathbf{Z}_j$$



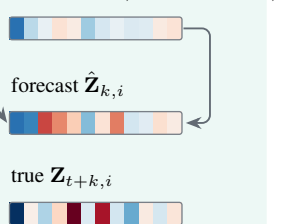
$$\text{gate } g_{k,i} = \sigma(\mathbf{w}^\top \mathbf{h}_{k,i}) = 0.99$$



strips: 384-D features, top 12 PCs

3 Correct, predict

$$\text{correction } \mathbf{r}_{k,i} = \mathbf{s} \odot \mathbf{W}_o \mathbf{h}_{k,i}$$



error: stay 1.65 $\rightarrow$ **ours 0.74**

all patches and steps in parallel